

Name: J. Renuka Devi

Reg.no: 192573007

Course code: DSA0602

Course name: Data Handling and Visualization

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Scenario 1: Student Academic Performance Dashboard

CODE:

```
# Import required libraries

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Read the CSV file

df = pd.read_csv("student_performance.csv")

# Display the dataset

print("Student Academic Performance Dataset")

print(df)

# Set the graph style

sns.set(style="whitegrid")

# Create scatter plot

plt.figure(figsize=(8,6))

sns.scatterplot(

    data=df,

    x="Attendance",

    y="Internal_Marks",

    hue="Department",

    s=120
```

)

Add title

```
plt.title("Student Academic Performance Dashboard")
```

Add labels

```
plt.xlabel("Attendance (%)")
```

```
plt.ylabel("Internal Marks")
```

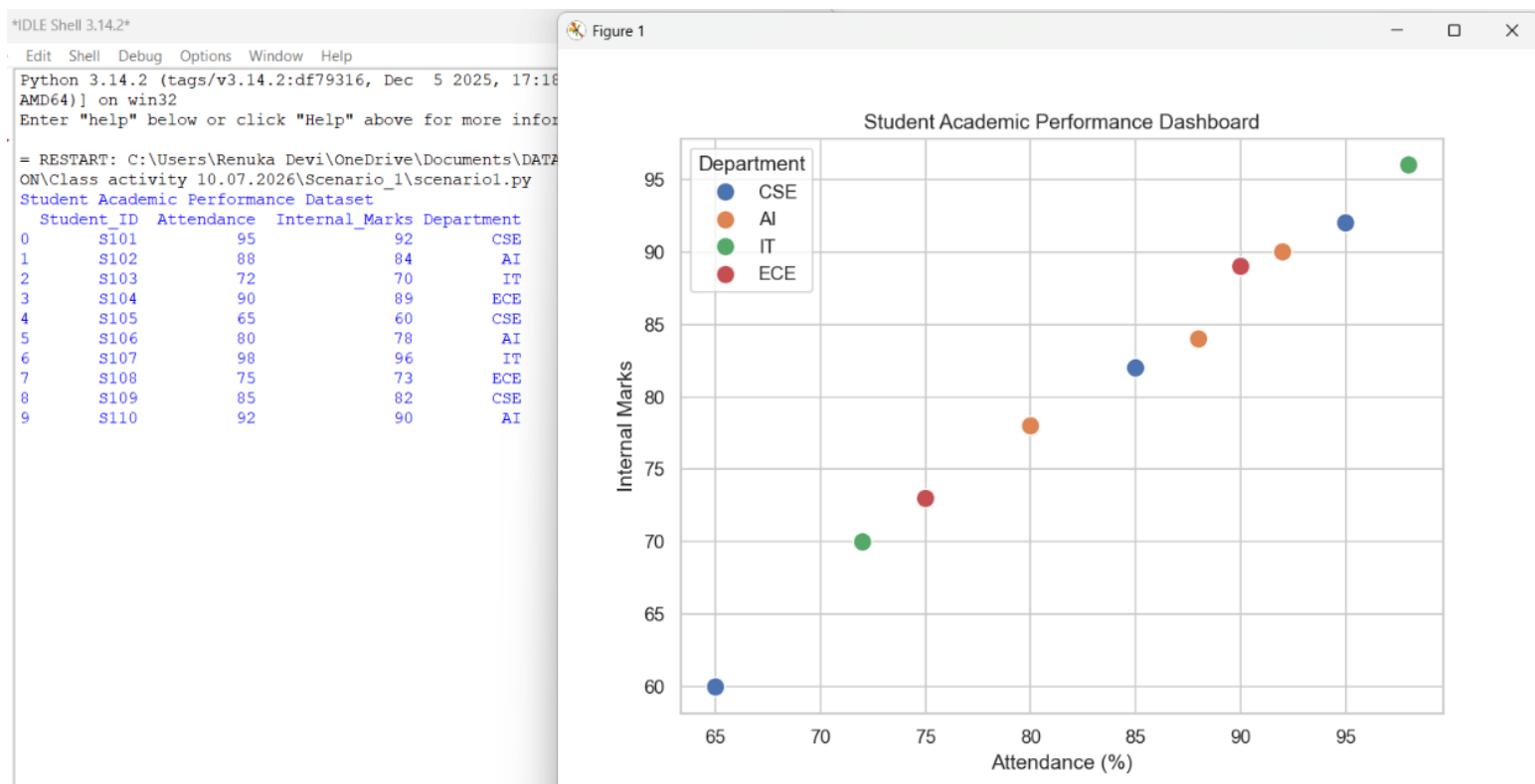
Display legend

```
plt.legend(title="Department")
```

Show graph

```
plt.show()
```

OUTPUT:



Scenario 2: Electric Vehicle Market Analysis

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt


# Read the dataset

df = pd.read_csv("electric_vehicle.csv")


# Assign colors to manufacturers

colors = {

    "Tata": "blue",

    "MG": "green",

    "BYD": "red",

    "Mahindra": "orange",

    "Hyundai": "purple",

    "Tesla": "black",

    "Kia": "brown",

    "Volvo": "pink"

}


plt.figure(figsize=(10,6))


# Create Bubble Chart

for company in df["Manufacturer"].unique():

    data = df[df["Manufacturer"] == company]

    plt.scatter(

        data["Battery_Capacity"],
```

```
data["Price"],  
s=data["Driving_Range"],  
color=colors[company],  
alpha=0.6,  
label=company  
)
```

```
# Add labels for each vehicle
```

```
for i in range(len(df)):
```

```
    plt.text(  
        df["Battery_Capacity"][i],  
        df["Price"][i],  
        df["Vehicle_Model"][i],  
        fontsize=8,  
        ha="left"  
    )
```

```
# Title and labels
```

```
plt.title("Electric Vehicle Market Analysis")
```

```
plt.xlabel("Battery Capacity (kWh)")
```

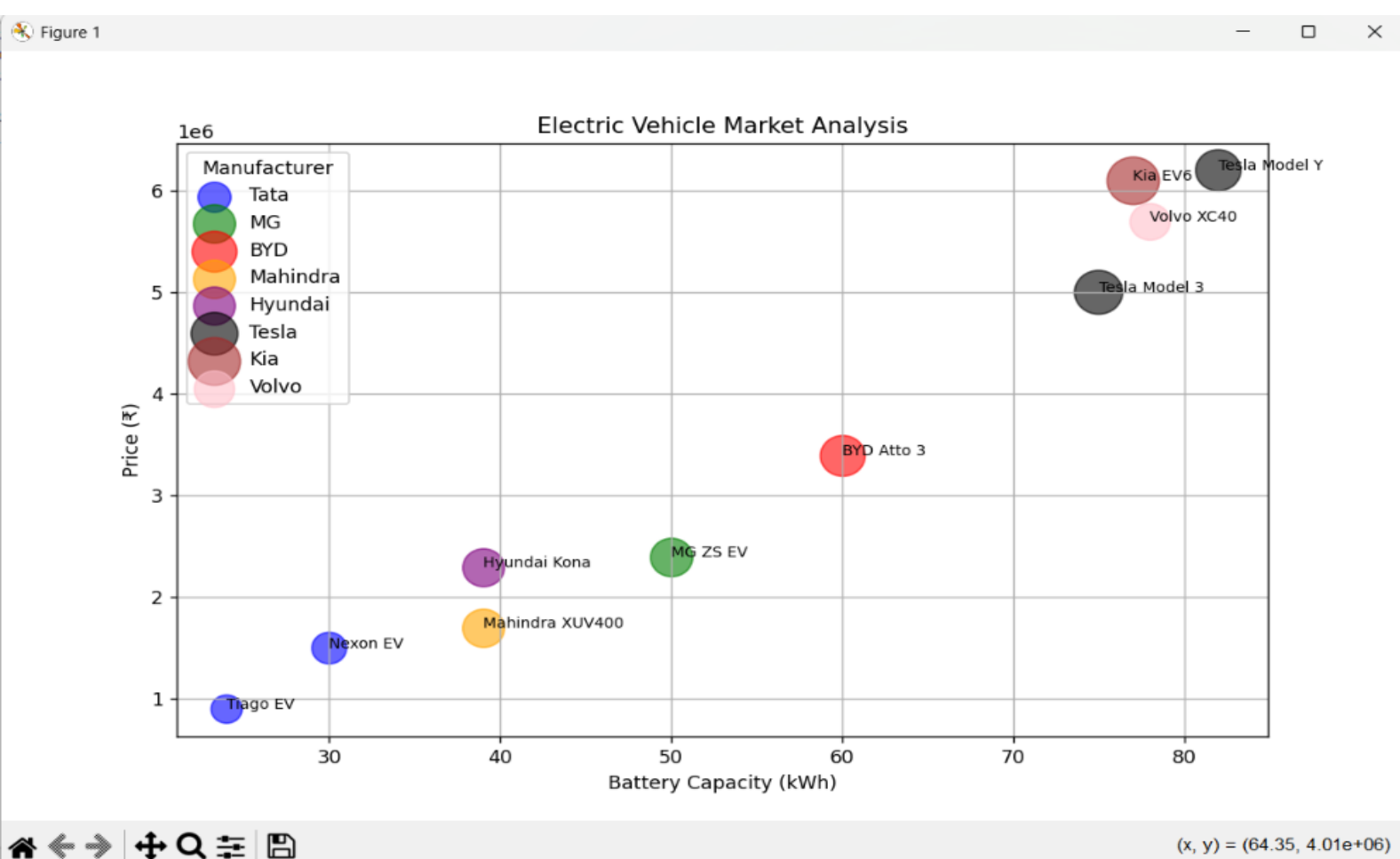
```
plt.ylabel("Price (₹)")
```

```
plt.legend(title="Manufacturer")
```

```
plt.grid(True)
```

```
plt.show()
```

OUTPUT:



Scenario 3: Hospital Patient Admission Analysis

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file
df = pd.read_csv("hospital_patient_admission.csv")

# Display the dataset
print(df)
```

```
# Colors for each department
colors = ["blue", "green", "orange", "purple", "red", "cyan"]

# Create bar chart
plt.figure(figsize=(10,6))

bars = plt.bar(
    df["Department"],
    df["Number_of_Patients"],
    color=colors
)

# Display treatment cost as labels on each bar
for bar, cost in zip(bars, df["Average_Treatment_Cost"]):
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 2,
        f"₹{cost}",
        ha='center',
        fontsize=10,
        fontweight='bold'
    )

# Title and labels
plt.title("Hospital Patient Admission Analysis", fontsize=16, pad=20)
plt.xlabel("Department")
plt.ylabel("Number of Patients")
```

```
# Rotate x-axis labels for better readability
```

```
plt.xticks(rotation=15)
```

```
# Grid
```

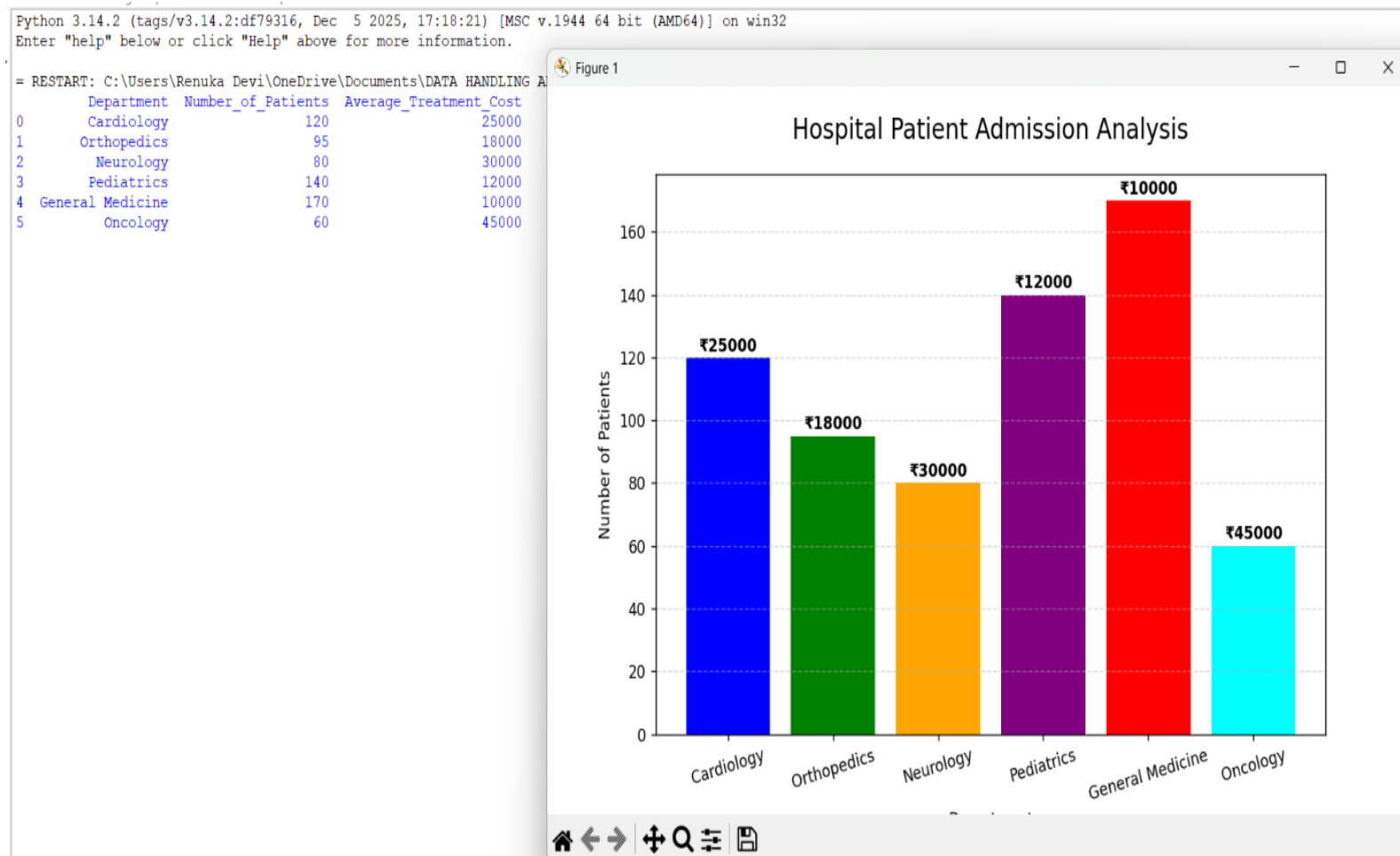
```
plt.grid(axis='y', linestyle='--', alpha=0.5)
```

```
# Show graph
```

```
plt.show()
```

```
plt.tight_layout()
```

OUTPUT:



Scenario 4: Weather Monitoring System

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt


# Read the dataset

df = pd.read_csv("weather_monitoring.csv")


# Display dataset

print(df)


# Create figure

plt.figure(figsize=(10,6))


# City-wise line styles and colors

styles = {

    "Chennai": ("blue", "-"),

    "Bangalore": ("green", "--"),

    "Delhi": ("red", ":")

}


# Draw line for each city

for city in df["City"].unique():

    city_data = df[df["City"] == city]
```

```
plt.plot(  
    city_data["Date"],  
    city_data["Temperature"],  
    color=styles[city][0],  
    linestyle=styles[city][1],  
    marker="o",  
    linewidth=2,  
    label=city  
)
```

Title and labels

```
plt.title("Weather Monitoring System", fontsize=16, pad=20)
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Temperature (°C)")
```

Legend

```
plt.legend(title="City")
```

Grid

```
plt.grid(True, linestyle="--", alpha=0.5)
```

Adjust layout

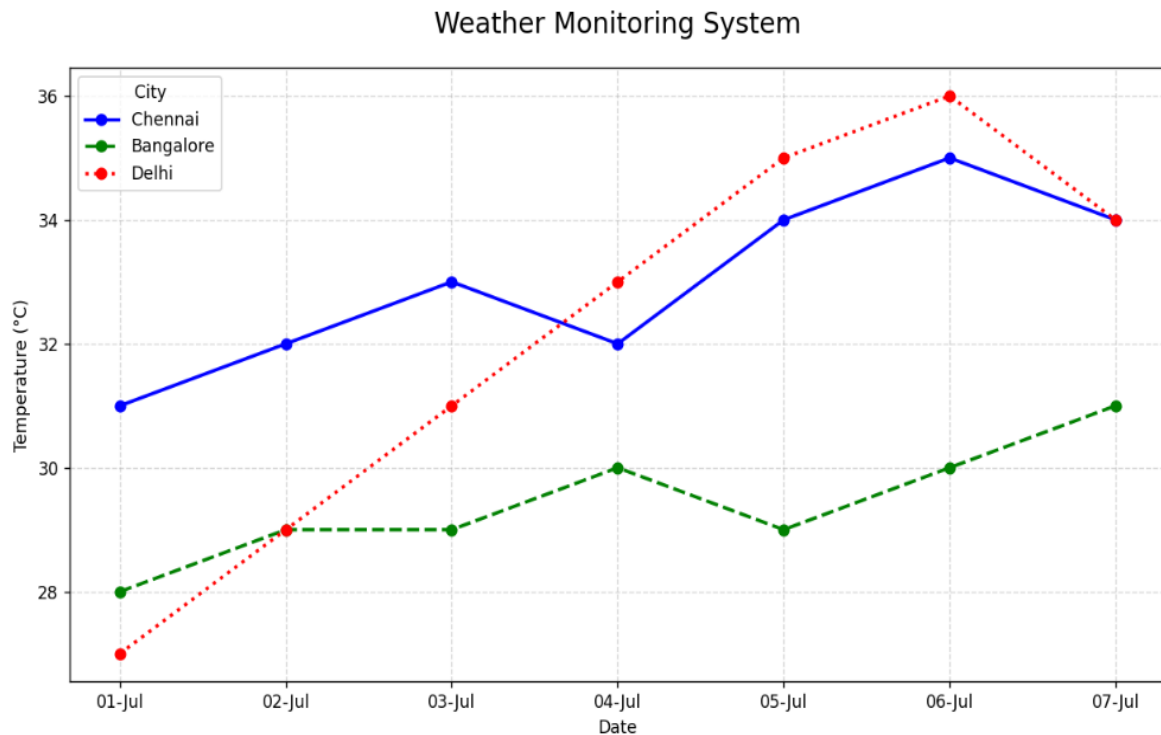
```
plt.tight_layout()
```

```
# Show graph
```

```
plt.show()
```

OUTPUT:

```
===== RESTART: C:\Python37\Python
0  01-Jul  31  Chennai
1  02-Jul  32  Chennai
2  03-Jul  33  Chennai
3  04-Jul  32  Chennai
4  05-Jul  34  Chennai
5  06-Jul  35  Chennai
6  07-Jul  34  Chennai
7  01-Jul  28  Bangalore
8  02-Jul  29  Bangalore
9  03-Jul  29  Bangalore
10 04-Jul  30  Bangalore
11 05-Jul  29  Bangalore
12 06-Jul  30  Bangalore
13 07-Jul  31  Bangalore
14 01-Jul  27  Delhi
15 02-Jul  29  Delhi
16 03-Jul  31  Delhi
17 04-Jul  33  Delhi
18 05-Jul  35  Delhi
19 06-Jul  36  Delhi
20 07-Jul  34  Delhi
```



Scenario 5: Supermarket Sales Analysis

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read dataset
```

```
df = pd.read_csv("supermarket_sales.csv")
```

```
print(df)
```

```
# Pivot data for grouped bar chart
```

```
pivot = df.pivot(index="Product_Category",  
                  columns="Branch",
```

```
values="Sales_Amount")
```

```
# Plot grouped bar chart
```

```
pivot.plot(kind="bar", figsize=(10,6))
```

```
plt.title("Supermarket Sales Analysis", fontsize=16, pad=20)
```

```
plt.xlabel("Product Category")
```

```
plt.ylabel("Sales Amount (₹)")
```

```
plt.legend(title="Branch")
```

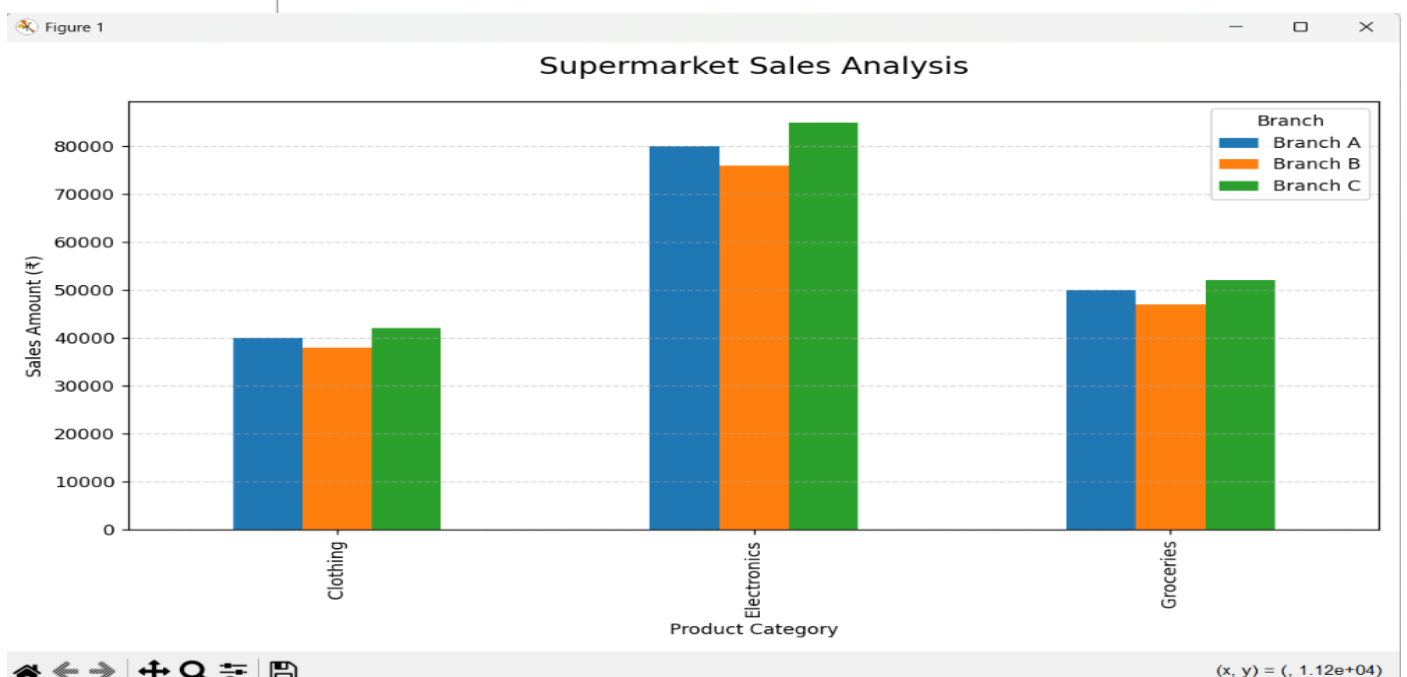
```
plt.grid(axis="y", linestyle="--", alpha=0.5)
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:

```
===== RESTART: C:\Users\Renuka Devi
  Month Product_Category Sales_Amount Branch
0 January      Groceries      50000 Branch A
1 January      Groceries      47000 Branch B
2 January      Groceries      52000 Branch C
3 January    Electronics      80000 Branch A
4 January    Electronics      76000 Branch B
5 January    Electronics      85000 Branch C
6 January      Clothing      40000 Branch A
7 January      Clothing      38000 Branch B
8 January      Clothing      42000 Branch C
```



Scenario 6: Employee Salary Distribution

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read the dataset

df = pd.read_csv("employee_salary.csv")

# Display dataset

print(df)

# Create Histogram

plt.figure(figsize=(10,6))

plt.hist(

    df["Salary"],

    bins=8,          # Try changing to 5, 10, or 12

    color="skyblue",

    edgecolor="black"

)

# Title and labels

plt.title("Employee Salary Distribution", fontsize=16, pad=20)

plt.xlabel("Salary (₹)")

plt.ylabel("Number of Employees")

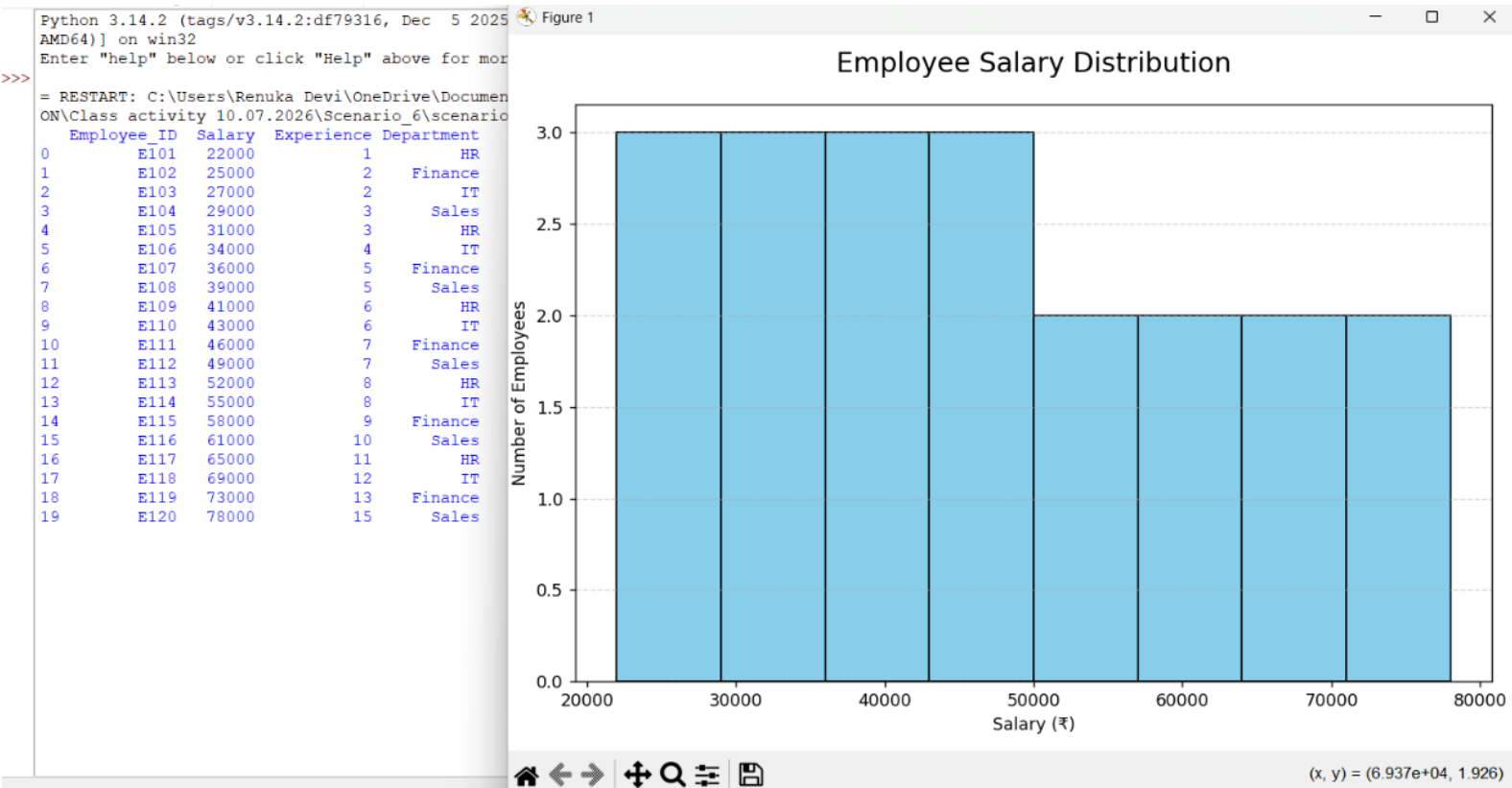
# Grid

plt.grid(axis='y', linestyle='--', alpha=0.5)
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



Scenario 7: Smartphone Market Comparison

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("smartphone_market.csv")
```

```
markers = {
```

```
    "Samsung": "o",
```

```
    "Apple": "s",
```

```
    "OnePlus": "^",
```

```
"Xiaomi": "D",  
"Realme": "*" }  
}
```

```
plt.figure(figsize=(10,6))
```

```
for brand in df["Brand"].unique():
```

```
    data = df[df["Brand"] == brand]
```

```
    plt.scatter(  
        data["RAM"],  
        data["Price"],  
        s=data["Storage"],  
        c=data["Processor_Speed"],  
        cmap="viridis",  
        marker=markers[brand],  
        alpha=0.8,  
        label=brand  
    )
```

```
plt.colorbar(label="Processor Speed (GHz)")
```

```
plt.title("Smartphone Market Comparison", fontsize=16, pad=20)
```

```
plt.xlabel("RAM (GB)")
```

```
plt.ylabel("Price (₹)")
```

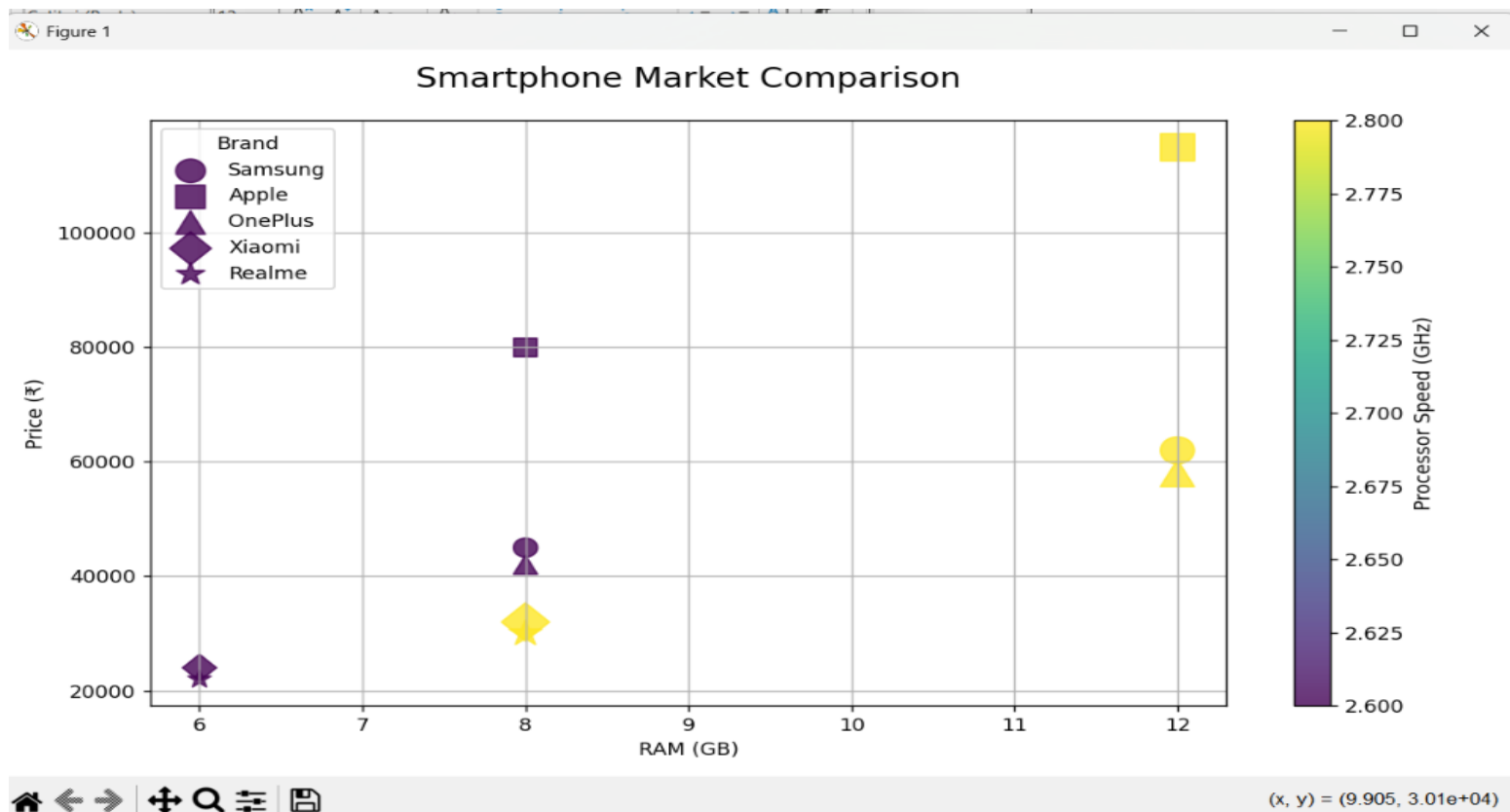
```
plt.legend(title="Brand")
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



Scenario 8: Online Movie Rating Analysis

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Read dataset
df = pd.read_csv("movie_rating.csv")

# Create pivot table
heatmap_data = df.pivot(
    index="Release_Year",
    columns="Genre",
    values="Average_Rating"
)
```

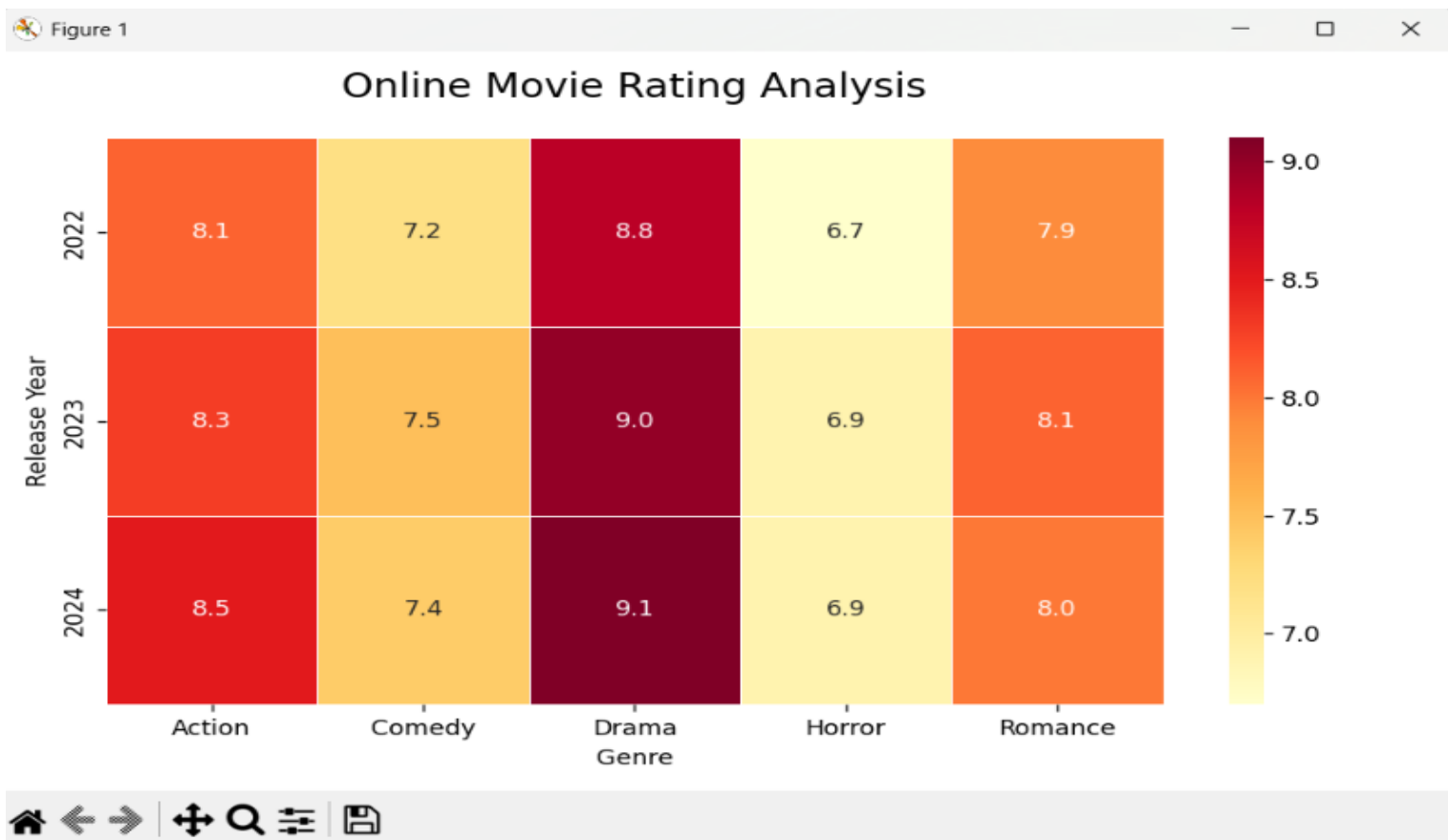
```

# Plot heatmap
plt.figure(figsize=(8,5))
sns.heatmap(
    heatmap_data,
    annot=True,
    cmap="YlOrRd",
    linewidths=0.5,
    fmt=".1f"
)

plt.title("Online Movie Rating Analysis", fontsize=16, pad=20)
plt.xlabel("Genre")
plt.ylabel("Release Year")
plt.tight_layout()
plt.show()

```

OUTPUT:



Scenario 9: COVID-19 State-wise Analysis

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("covid_state.csv")

# Sort states by active cases

df = df.sort_values(by="Active_Cases", ascending=False)

plt.figure(figsize=(10,6))

bars = plt.bar(
    df["State"],
    df["Active_Cases"],
    color=plt.cm.Reds(df["Active_Cases"] / df["Active_Cases"].max())
)

# Add value labels

for bar in bars:
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 100,
        str(int(bar.get_height())),
        ha='center',
        fontsize=9
    )
```

```
plt.title("COVID-19 State-wise Active Cases", fontsize=16, pad=20)
```

```
plt.xlabel("State")
```

```
plt.ylabel("Active Cases")
```

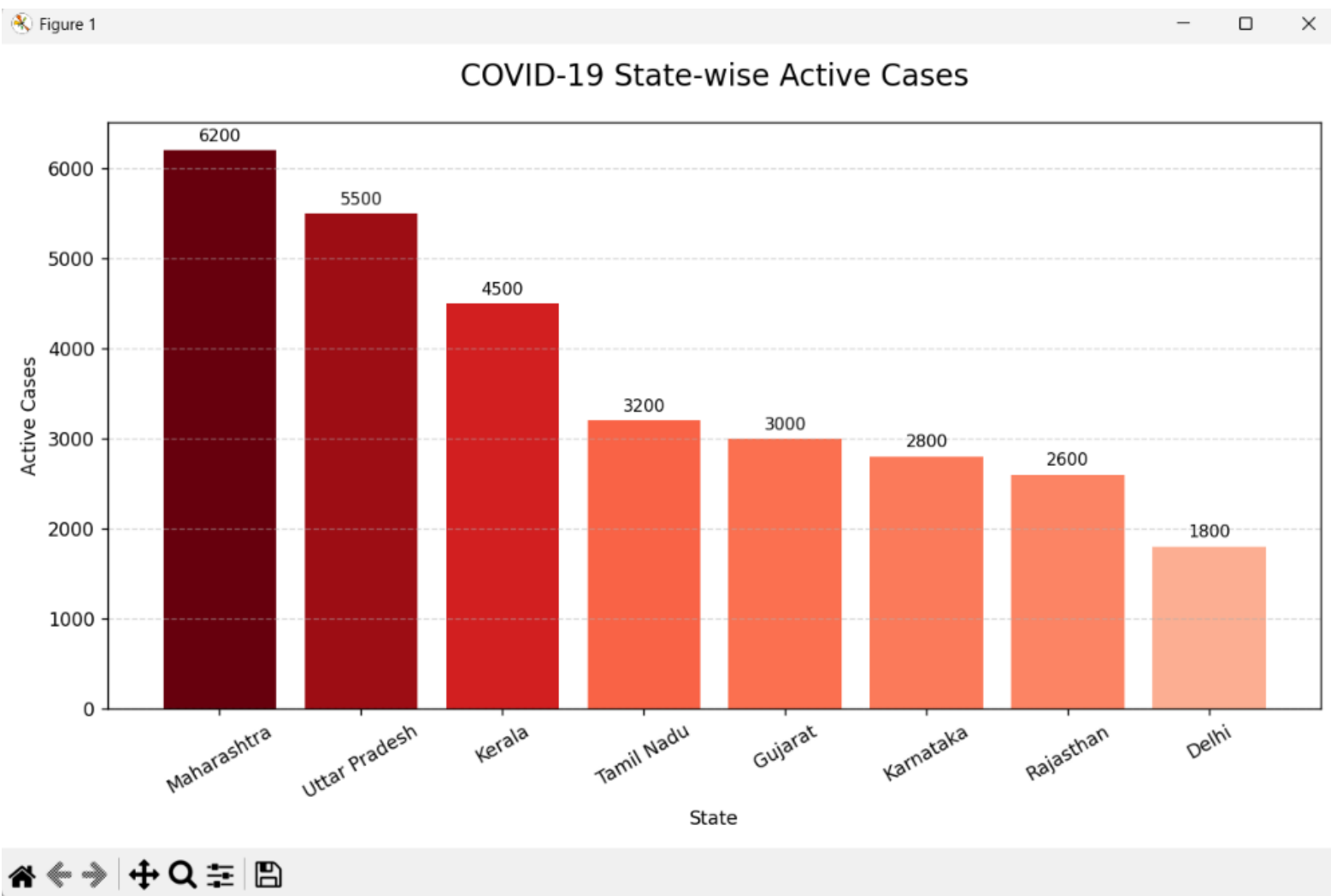
```
plt.xticks(rotation=30)
```

```
plt.grid(axis="y", linestyle="--", alpha=0.5)
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("iris_dataset.csv")

# Colors for each species

colors = {

    "setosa": "red",

    "versicolor": "green",

    "virginica": "blue"

}

# Shapes for each species

markers = {

    "setosa": "o",

    "versicolor": "s",

    "virginica": "^"

}

plt.figure(figsize=(10,6))

for species in df["Species"].unique():

    data = df[df["Species"] == species]

    plt.scatter(

        data["Sepal_Length"],

        data["Petal_Length"],

        s=data["Sepal_Width"] * 80, # Point size

        color=colors[species],      # Color = Species

        marker=markers[species],    # Shape = Species

        edgecolors="black",
```

```

        label=species
    )
plt.title("Iris Flower Classification", fontsize=16, pad=20)
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.legend(title="Species")
plt.grid(True)
plt.tight_layout()
plt.show()

```

OUTPUT:

